Experiment 5(b):Modelling and simulation of a 4-Layer IoE Architecture (Perception–Network–Middleware–Application) using Node-

RED, MQTT and a Cloud Dashboard

Objectives

1. To understand the four-layer IoE architecture.
2. To simulate sensor data generation in the Perception layer.
3. To transmit sensor data using MQTT in the Network layer.
4. To process and analyze sensor data using Node-RED in the Middleware layer.
5. To visualize real-time sensor information using a Cloud Dashboard.

Software Required

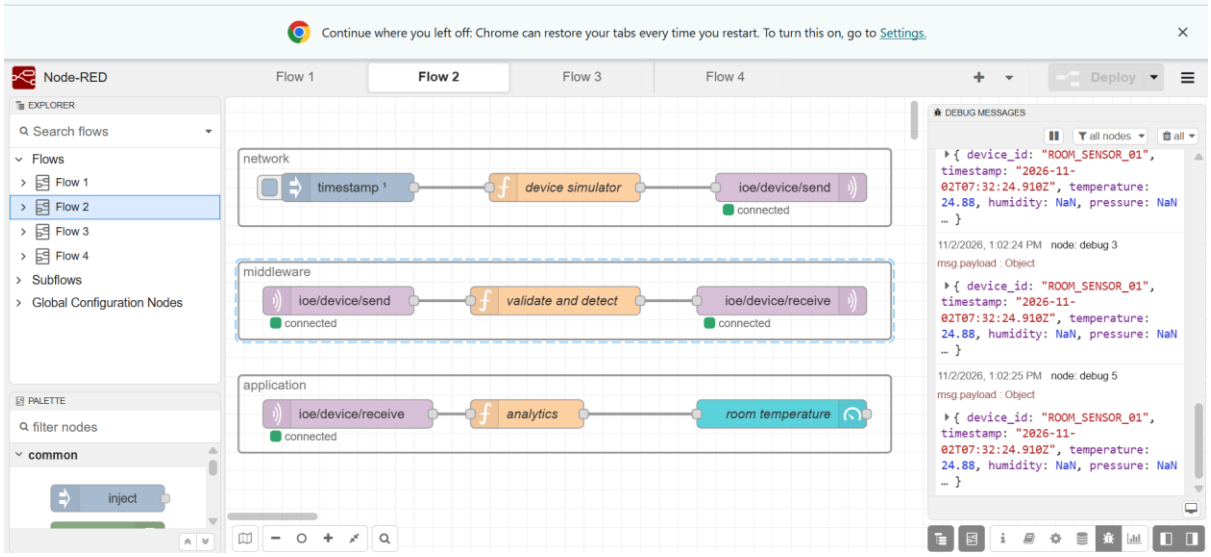
- Node-RED
- MQTT broker (Mosquitto or Node-RED MQTT nodes)
- Node-RED Dashboard

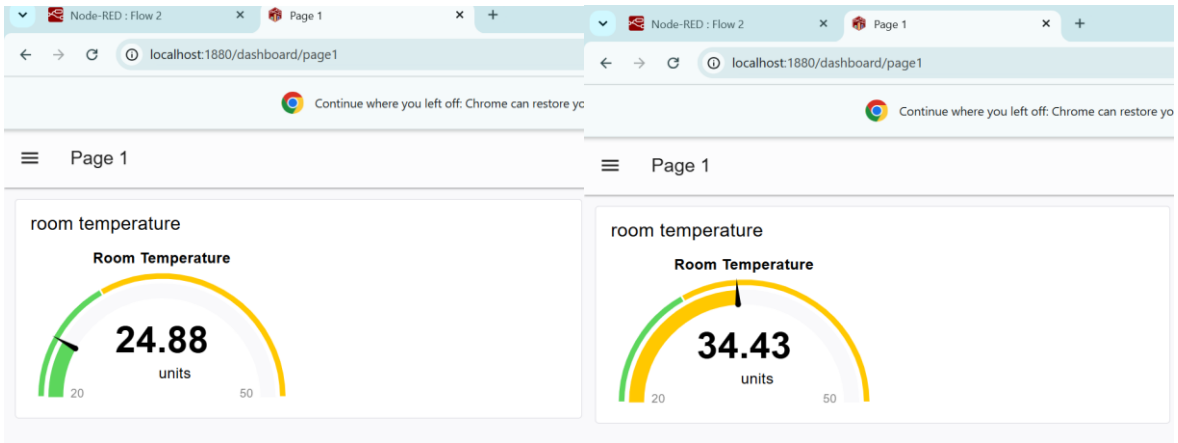
Procedure

1. Open Node-RED and create a new flow.
2. Add an Inject node to simulate sensor readings.
3. Add a Function node to generate temperature and humidity values.
4. Configure an MQTT Output node.
5. Set the MQTT topic as ioe/sensor/data.
6. Connect the MQTT Output node to the MQTT broker.
7. Add an MQTT Input node to receive sensor information.
8. Connect a Function node to validate and process the received data.
9. Add Debug nodes to monitor the output.
10. Create a dashboard using gauges, charts and text displays.

11. Deploy the flow and monitor the real-time sensor data.

Output:





Output Table

S.No.	Parameter	Expected Output
1	Temperature Sensor	Sensor data generated
2	MQTT topic	ioe/device/send
3	Data Transmission	Successful
4	Data processing	Temperature validated
5	Dashboard	Room temperature displayed (24.88 °C)

Result

The four-layer IoE architecture was successfully modelled and simulated using Node-RED and MQTT. Temperature sensor data was generated, transmitted, processed and displayed on the dashboard with a reading of 24.88 °C.